

Contact

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Top Skills

Large Language Models (LLM)

Lean 4

Research Skills

Certifications

Mindshift: Break Through Obstacles to Learning and Discover Your Hidden Potential (with Honors)

Learning How to Learn: Powerful mental tools to help you master tough subjects (with Honors)

Honors-Awards

Best Strategy Report

SURGE Fellow

Sloan Scholar, Alfred P. Sloan Foundation's Minority Ph.D. (MPHD) Program, awarded in 2018-2019

Saburo Muroga Fellowship, Computer Science Excellence Fellowship

Most Cited Paper Certificate

Publications

Why and when can deep-but not shallow-networks avoid the curse of dimensionality: A review

Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?

High-performance and scalable on-chip digital Fourier transform spectroscopy

Are emergent abilities of Large Language Models a mirage?

A surprising linear relationship predicts test performance in deep networks

Brando Miranda

Computer Science Graduate Student at SAIL
Stanford, California, United States

Summary

Brando Miranda is currently a Computer Science AI/ML Ph.D. student at Stanford University under the supervision of Professor Sanmi Koyejo in the Stanford Trustworthy AI Research (STAIR) group. Previously, Miranda was a graduate researcher at University of Illinois Urbana-Champaign, a Research Assistant at the Massachusetts Institute of Technology (MIT)'s Center for Brain Minds and Machines (CBMM), and a graduate student at MIT. Miranda's research interests focuses in data-centric machine learning for Frontier Models (FMs), Transformative Artificial Intelligence (TAI), and machine learning for mathematics and verified code. Miranda earned his Master of Engineering in Electrical Engineering and Computer Science at MIT, conducting research on Deep Learning Theory under the guidance of Professor Tomaso Poggio. Miranda has been the recipient of several awards; including the prestigious NeurIPS Outstanding Main Track Paper Award (top 0.4% and only 2 papers selected), the Most Cited Paper Certificate awarded by International Journal of Automation & Computing (IJAC), two Honorable Mention with the Ford Foundation Fellowship, Computer Science Excellence Saburo Muroga Endowed Fellow, Stanford School of Engineering fellowship, and is currently an EDGE Scholar at Stanford University. Miranda is more than a researcher: he is an innovator, a communicator, and deeply passionate about the future. Therefore, he collaborated and worked as a Machine Learning advisor at the AI start up Morph Labs. He was a key contributor to the Morph Prover, a frontier model for mathematics and verified code in Lean 4, and MoogLe.ai, a search engine for verified code in Lean 4. Aligned with his passion for innovation and start-ups, he has also advised Wise, a Stanford based startup focused on transforming sales performance with AI.

Website: <https://brando90.github.io/brandomiranda/>

Twitter: <https://x.com/BrandoHablando>

Experience

Stanford University

PHD Graduate Student

September 2022 - Present (3 years 9 months)

Stanford, California, United States

PhD in Machine Learning.

Amazon Web Services (AWS)

Applied Scientist Intern

June 2024 - September 2024 (4 months)

Cupertino, California, United States

Applied Scientist intern working on LLMs for the Lean 4 Proof Assistant and Programming Language for formal verification.

Morph Labs

Machine Learning Consultant @ Morph Labs

October 2023 - December 2023 (3 months)

San Francisco Bay Area

- Consulted for ML Start Up

- Key contributor to develop the model [Morph Prover](<https://huggingface.co/morph-labs/morph-prover-v0-7b>) frontier model for mathematics and formal verification (Lean 4)

- Contributed to the development of [MoogLe.ai](<https://www.moogLe.ai/>), search engine for mathematical theorems in Lean 4

IBM

Machine Learning Research Intern

May 2022 - August 2022 (4 months)

Yorktown Heights, New York, United States

Doing research in program synthesis with machine learning

University of Illinois at Urbana-Champaign

3 years 9 months

PHD Graduate Student

September 2018 - May 2022 (3 years 9 months)

Urbana-Champaign, Illinois Area

PhD in Machine Learning.

CS446/ECE449: Machine Learning (Spring 2021)

January 2021 - May 2021 (5 months)

IBM

Research Intern

May 2021 - August 2021 (4 months)

Massachusetts Institute of Technology

3 years

Research Assistant at CBMM

2016 - August 2018 (2 years)

Cambridge, MA

Statistical Learning Theory and Applications (9.520/6.860) Teaching Assistant

September 2016 - December 2016 (4 months)

Cambridge, MA

Instructors: Tomaso Poggio, Lorenzo Rosasco

- Helped prepare slides for class and OCW
- Helped grade final project papers and presentation
- Provided extra assistance for office hours if needed
- The topics range from: Reproducing Kernel Hilbert Spaces, Tikhonov Regularization, Logistic Regression and SVMs, Regularized Least Squares, Iterative Regularization via Early Stopping, Large Scale Kernel Methods, Sparsity based Regularization, Convex Relaxation and Proximal Gradient, Structured Sparsity, Multiple Kernel Learning, Online Learning, Manifold Regularization, Multi-Output learning, Learning Theory, Generalization Error and Stability, Dictionary Learning, Approximation Theory for shallow and deep networks, Learning data representation for invariance and selectivity.

Introduction to Algorithms (6.006) Teaching Assistant

February 2016 - May 2016 (4 months)

Cambridge, MA

Instructors: Nancy Lynch, Bruce Tidor and Aleksander Madry.

- Instructed students across bi-weekly recitations
- Held weekly Office Hours
- Helped prepare problem sets and exams
- Graded exams
- The topics range from: algorithmic thinking, Divide and Conquer, asymptotic and recursion, Sorting, data structures e.g. Heaps, Binary Search Trees (BSTs), AVL trees, hashing, graph algorithms, depth first search (DFS), breadth first search (BFS), shortest paths, Dijkstra's algorithm, Bellman-Ford, Iterative Algorithms Gradient Descent, Newton's Method, Dynamic Programming.

Design and Analysis of Algorithms (6.046/18.410) Teaching Assistant
September 2015 - December 2015 (4 months)
Cambridge, MA

Instructors: Shafi Goldwasser, Dana Moshkovitz, Nir N. Shavit

- Instructed students across weekly recitations
- Held weekly Office Hours
- Helped prepare problem sets and exams
- The topics range from: divide-and-conquer, FFT, randomization, monte Carlos, las vegas, advanced data structures, Van Emde Boas Trees, B-trees, dynamic programming, greedy algorithms, minimum spanning trees, incremental improvement, network flows, complexity theory, NP-completeness and reductions, linear programming, approximation algorithms, number theoretic algorithms, cryptography, sublinear algorithms, parallel algorithms and amortized analysis,

Massachusetts Institute of Technology (MIT)
Introduction to Machine Learning (6.036) Teaching Assistant
February 2015 - May 2015 (4 months)
Cambridge, MA

Instructors: Tommi S Jaakkola, Suvrit Sra and Regina Barzilay

- Instructed students across bi-weekly recitations
- Held weekly Office Hours
- Helped prepare problem sets, projects and exams
- Graded problem sets, projects and exams
- The topics range from: perceptron algorithm, linear regression, clustering, collaborative filtering, SVMs, kernel methods, Boosting, neural networks, autoencoders, intro to statistical learning theory and generalization, mixture models, the EM algorithm, Bayesian networks, hidden markov models (HMM) and reinforcement learning.

Massachusetts Institute of Technology (MIT)
Mathematics for Computer Science (6.042/18.062) Teaching Assistant
September 2014 - December 2014 (4 months)
Cambridge, MA

Instructors: F. Thomson Leighton and Ankur Moitra

- Instructed 32 students across two recitations
- Held 4 recitations and problem solving sessions weekly
- Held weekly Office Hours
- Helped prepare problem sets and exams
- Graded exams

- Taught concepts in discrete mathematics with direct application to Computer Science
- The topics range from: proof techniques, number theory, graph theory, recurrences, asymptotics, introduction to combinatorics, probability and statistics

Rackspace, the #1 managed cloud company

Software Engineering Intern

June 2014 - August 2014 (3 months)

- Built a proxy to extend the functionality of the Redis Database.
- Implemented a concurrent proxy.
- Added a proxy security functionality.
- Presented and delivered a demo and presentation to engineering team and higher level management.

Adobe

Software Engineering Intern

June 2013 - August 2013 (3 months)

San Jose, California

- Project: optimized Machine Learning software for sentiment analysis in Spanish.
- Researched different machine learning and natural language processing (NLP) techniques to optimize the Naïve Bayes algorithm.
- Developed Natural Language Processing (NLP) techniques to deal with negation in sentences.
- Developed different methods for automatic extraction of training data.
- Presented and delivered a demo to Adobe's engineering department.

Massachusetts Institute of Technology (MIT)

Undergraduate Research Assistant (UROP)

June 2012 - August 2012 (3 months)

Cambridge, Massachusetts

Project: Crowd-Sourcing the Design of Drug-Delivery Vehicles for Cancer Treatment.

- Developed website user interface (UI) that enables the general population to design drug delivery vehicles for cancer treatment.
- Built the communication network allowing reliable interaction amongst the user interface, database and back-end of the project.
- Presented and explained the project to the research lab.

Greengates School Mexico

Student

1995 - September 2010 (15 years)

Education

Stanford University

Doctor of Philosophy - PhD, Machine Learning · (September 2022 - 2025)

University of Illinois Urbana-Champaign

Doctor of Philosophy - PhD, Artificial Intelligence · (2018 - 2022)

Massachusetts Institute of Technology

Master's Degree, Machine Learning · (2014 - 2016)

Massachusetts Institute of Technology

Bachelor's Degree, Electrical Engineering and Computer
Science · (2010 - 2014)

Greengates School Mexico

International Baccalaureate Diploma · (1995 - 2010)